

QUEEN ALIA INTL, Jordan

WMO: 402720

Lat: 31.723N

Lon: 35.993E

Elev: 730

StdP: 92.86

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	0.9	-11.2	1.6	10.9	-8.7	2.0	11.8	14.8	7.8	13.1	7.1	1.2	40	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	37.1	19.5	35.8	19.0	34.2	18.7	22.7	32.1	21.7	31.3	20.9	30.3	5.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	16.0	26.4	18.8	14.9	26.1	17.8	14.0	25.4	70.9	31.8	66.9	31.5	63.8	30.2	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.8	8.7	DB	-3.1	40.3	1.9	1.5	-4.4	41.4	-5.5	42.2	-6.6	43.1	-8.0	44.2	(4)
(5)			WB	-4.2	24.2	2.2	2.4	-5.7	25.9	-7.0	27.3	-8.2	28.6	-9.8	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	8.0	9.3	12.1	16.1	20.5	23.0	24.9	24.9	23.0	19.6	14.0	9.6	(6)	
	DBStd	6.81	2.52	3.04	3.77	4.09	3.63	2.43	2.02	2.03	2.37	3.20	3.28	2.80	(7)	
	HDD10.0	183	72	45	18	2	0	0	0	0	0	0	5	41	(8)	
	HDD18.3	1313	321	254	198	93	18	0	0	0	0	22	134	272	(9)	
	CDD10.0	2786	9	24	83	184	326	391	462	463	391	299	125	28	(10)	
	CDD18.3	874	0	0	4	26	86	142	204	204	142	62	5	0	(11)	
	CDH23.3	12439	0	6	95	463	1387	2130	2848	2756	1833	832	85	5	(12)	
(13)	CDH26.7	6165	0	1	25	161	636	1081	1561	1503	874	309	14	1	(13)	
(14) Wind	WSAvg	3.4	3.5	3.7	3.7	3.6	3.7	3.8	4.1	3.5	2.9	2.5	3.0	3.2	(14)	
(15) Precipitation	PrecAvg	222	58	53	31	9	2	0	0	0	0	5	22	41	(15)	
	PrecMax	402	115	190	87	56	20	1	0	0	1	22	102	136	(16)	
	PrecMin	86	2	12	5	0	0	0	0	0	0	0	1	0	(17)	
	PrecStd	68	26	35	23	12	4	0	0	0	0	6	24	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	24.1	29.8	33.3	37.1	38.1	39.9	39.1	37.8	34.2	28.7	23.8	(19)	
		MCWB	10.9	11.6	14.2	15.6	18.2	18.5	20.7	20.7	19.2	17.1	14.2	11.2	(20)	
	2%	DB	17.9	21.0	25.9	30.8	34.7	36.0	36.9	36.2	35.0	32.1	25.9	20.0	(21)	
		MCWB	9.5	10.5	12.9	14.6	17.2	17.9	20.6	20.0	18.4	16.9	13.4	10.6	(22)	
	5%	DB	15.8	18.1	23.1	28.2	32.6	34.1	35.0	34.9	33.1	30.1	23.2	17.4	(23)	
		MCWB	9.0	9.6	11.9	13.7	16.5	17.9	19.3	19.9	18.5	16.7	12.8	9.8	(24)	
	10%	DB	13.8	15.9	20.2	25.9	30.2	32.2	33.3	33.2	31.3	28.0	21.2	15.7	(25)	
		MCWB	8.5	8.9	10.7	13.0	15.9	17.7	19.0	19.4	18.4	16.1	12.3	9.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	13.3	15.6	17.4	21.3	21.7	24.3	24.2	23.2	21.4	16.7	13.6	(27)	
		MCDB	17.0	21.7	25.4	29.6	33.0	32.2	35.2	33.2	30.2	29.4	22.3	19.1	(28)	
	2%	WB	10.8	11.5	14.0	15.9	19.0	20.5	22.4	22.8	21.6	19.4	15.1	12.0	(29)	
		MCDB	15.4	17.9	23.1	26.4	31.0	31.2	32.6	32.2	29.9	26.8	21.6	16.6	(30)	
	5%	WB	9.8	10.5	12.7	14.7	17.6	19.6	21.3	21.9	20.5	18.2	14.1	11.0	(31)	
		MCDB	14.1	16.2	20.8	25.5	29.4	30.4	31.5	31.5	28.9	25.4	20.5	15.7	(32)	
	10%	WB	9.0	9.5	11.5	13.6	16.4	18.7	20.4	20.9	19.7	17.2	13.2	10.1	(33)	
		MCDB	12.8	14.6	18.7	24.1	27.8	29.5	30.3	30.1	28.1	25.1	19.7	14.1	(34)	
(35) Mean Daily Temperature Range	MDBR		10.8	11.6	13.5	15.8	17.0	17.7	17.2	17.2	17.0	15.8	13.6	11.8	(35)	
	5% DB	MCDBR	14.6	15.7	17.6	18.7	19.5	20.4	19.3	18.9	19.2	17.8	16.1	14.9	(36)	
		MCWBR	8.2	8.0	8.5	8.4	8.7	9.1	8.5	7.8	7.9	8.3	8.4	8.4	(37)	
	5% WB	MCDBR	12.5	13.4	15.6	17.1	18.0	18.4	17.7	17.5	17.1	15.6	13.7	12.4	(38)	
		MCWBR	7.9	7.7	8.3	8.4	9.3	9.2	8.8	8.3	8.1	8.3	8.0	7.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.376	0.422	0.469	0.468	0.394	0.400	0.407	0.403	0.422	0.369	0.347	(40)	
	taud		2.254	2.153	2.005	1.888	1.918	2.161	2.154	2.138	2.146	2.076	2.208	2.257	(41)	
	Ebn at Noon		864	870	853	828	830	890	882	872	862	808	827	838	(42)	
	Edh at Noon		111	135	169	198	194	152	152	152	145	144	116	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.17	3.98	5.37	6.51	7.53	8.34	8.09	7.43	6.36	4.87	3.66	2.97	(44)	
	RadStd		0.18	0.27	0.29	0.26	0.30	0.11	0.10	0.10	0.16	0.18	0.18	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.48	N/A	-2.51	N/A	N/A	N/A	N/A	N/A	+160	+107	(46)	
(47)	Regional (18 neighbors)	+0.44	N/A	-1.95	N/A	N/A	N/A	N/A	-60	+140	+92	(47)	

Nomenclature: See separate page